AMBROSIA Platform

Dashboard Views & Chart Configuration

Implementation Specification

Based on Stakeholder Thematic Analysis (DR1–DR5)

Addressing: View Definitions • Chart–View Mapping • Per-View Chart Configuration

Introduction

This document provides the conceptual structure and configuration inputs needed to wire the AMBROSIA dashboard's multi-view system. It directly addresses the three implementation questions (i) view/mode definitions, (ii) view-to-chart mapping, and (iii) per-view chart representation, grounding every decision in the stakeholder feedback collected across the DRs and ISGs.

The platform currently has seven implemented risk charts:

- C1 – Toxin Concentration vs. Time
- C2 – Pathogen Concentration vs. Time (3D view)
- C3 – Probability of Illness vs. Time
- C4 – Cases per 100k vs. Time
- C5 – Seasonal Heatmap (baseline comparison)
- C6 – Geographic Heatmap
- C7 – Climate-Adjusted Scenarios vs. Time

These charts serve as the reference catalogue based on the MVP dashboard . No new chart definitions are proposed. The recommendations below focus exclusively on how to organise, configure, and present these existing charts across role-specific views.

1. View / Mode Definitions

Based on the five stakeholder categories surveyed and their distinct needs, we recommend six dashboard views, although the technician view can isn't conceder important at this stage . Each view is justified by direct stakeholder feedback themes.

1.1 Default View

Purpose: A balanced, general-purpose overview for first-time users or those who have not yet selected a role. Provides a broad orientation to all risk dimensions without assuming a specific workflow.

Stakeholder justification: Multiple groups across rounds noted the importance of a clear, quick overview of where risks are highest (Policy Actors DR1), and dashboard that provide a clear overview for strategic decision-making (Producers DR1). The Default view satisfies the onboarding need before role customisation takes effect.

Conceptual focus: Broad geographic and temporal risk snapshot. All seven charts available in a standard configuration. No aggressive filtering or pre-selection. Serves as the baseline from which other views specialise.

1.2 Advisor View

Purpose: Supports agricultural advisors in evaluating infection risks across regions and seasons, preparing mitigation strategies, and generating evidence for policy-aligned recommendations.

Stakeholder justification: Advisors consistently requested visual tools (maps, graphs, bar charts) with risk prediction and probability of exceedances (DR2), contextualised insights combining quantitative data with expert opinions (DR5), and downloadable raw data for independent analysis (DR4). Their anticipated use centres on risk assessment, crisis management, and seasonal advisory (DR1, DR3, DR5).

Conceptual focus: Risk prediction and trend analysis. Multi-region comparison. Uncertainty indicators. Medium-to-long time horizons (seasonal to multi-year). Climate integration. Emphasis on data export capability.

1.3 Policy Actor View

Purpose: Enables regulatory professionals and policy planners to monitor risk evolution, perform risk classification, detect emerging trends, and generate evidence for policy decisions.

Stakeholder justification: Policy actors requested map-based visualisation with colour-coded intensity (DR1), policy-relevant outputs such as forecasting (that we consider difficult to be done and quit out of scopus for now) and trend detection (DR2), risk scoring and dashboards (DR4), combined food risk insights from a multi-risk perspective (DR5), and causal explanations of risk interrelations (DR5). Their use cases include risk classification, strategic oversight, scenario analysis, and regulatory enforcement (DR1, DR2, DR4, DR5).

Conceptual focus: Aggregated national/regional risk overview. Long time horizons (multi-season to multi-year trends). Multi-hazard comparisons. Epidemiological indicators (cases per 100k). Climate scenario analysis. Emphasis on monitoring, trend detection, and exportable summary outputs.

1.4 Producer View

Purpose: Supports food producers and farmers in operational risk assessment, harvest monitoring, supplier evaluation, and supply chain decision-making.

Stakeholder justification: Producers requested dashboards for strategic decision-making and supplier communication (DR1), granular statistical data including time histories, forecasts, and statistical tables (DR1), interactive visual tools with real-time alerts for extreme weather impacts (DR5), and chat functionality for inter-user exchange (DR1). Their use focuses on operational risk assessment, harvest monitoring, and proactive risk awareness (DR1, DR2).

Conceptual focus: Operationally granular. Short-to-medium time horizons (current season, next harvest window). Local geographic scope. Detailed statistical breakdowns. Real-time alert emphasis. Crop-specific and supply-chain-aware. Highest data density of all views.

1.5 Distributor View

Purpose: Provides distributors and traders with quality forecasting tools, sourcing risk insights, and contamination indicators relevant to supply chain safety and trading decisions.

Stakeholder justification: Distributors emphasised interactivity and usability (DR2), action-oriented outputs with practical recommendations (DR2), graphical risk classification of areas (DR3), and contamination risks in raw materials for supply chain safety (DR3). Their use centres on market-facing quality forecasting, feed production support, and food safety assurance (DR2, DR3, DR5).

Conceptual focus: Simplified, action-oriented. Medium time horizons (upcoming harvest/sourcing window). Regional geographic scope aligned with sourcing areas. Focus on contamination risk in commodities. Clean, interactive interface with minimal technical complexity.

1.6 Technician View

Purpose: Equips technicians and technical advisors with precise predictive tools, quantified risk indicators, storage monitoring data, and detailed climate-pathogen correlations for crop management decisions.

Stakeholder justification: Technicians requested traffic-light colour-coded alerts (DR3), predictive tools showing mycotoxin risk per campaign/area (DR3), quantified risk indicators including probabilities and crop loss estimates (DR5), storage risk monitoring with aeration/temperature alerts (DR5), geospatial context via satellite/environmental mapping (DR5), and risk-yield integration (DR3). Their use includes advisory system integration, alert dissemination across networks, and decision support for sowing and crop management (DR3).

Conceptual focus: Technically detailed and prediction-focused. Short-to-medium time horizons (campaign/season). Field-to-local geographic scope. Climate-pathogen correlation emphasis. Colour-coded alert system. High precision in climate forecasting. Storage and post-harvest dimension.

2. View → Chart Mapping

The following matrix maps each of the seven existing charts to the six views. Charts are classified as Primary (prominently displayed, high priority), Secondary (available but smaller or collapsed by default), or Not Included (—) for each view. The mapping is driven by the stakeholder feedback themes summarised in Section 1.

Chart	Default	Advisor	Policy Actor	Producer	Distributor	Technician
C1: Toxin Conc. vs. Time	Primary	Primary	Secondary	Primary	Primary	Primary
C2: Pathogen Conc. vs. Time (3D)	Secondary	Primary	Secondary	Primary	Secondary	Primary
C3: Probability of Illness vs. Time	Secondary	Primary	Primary	Secondary	Secondary	Primary
C4: Cases per 100k vs. Time	Secondary	Secondary	Primary	—	—	Secondary
C5: Seasonal Heatmap	Primary	Primary	Primary	Primary	Primary	Primary
C6: Geographic Heatmap	Primary	Primary	Primary	Primary	Primary	Primary
C7: Climate-Adj. Scenarios vs. Time	Secondary	Primary	Primary	Secondary	—	Primary

Legend: **Primary** = prominently displayed, high priority. **Secondary** = available but collapsed/smaller by default. — = not included in this view.

2.1 Common Charts (All Views)

Two charts appear as Primary across all six views:

- C5 – Seasonal Heatmap: Universally valued. Addresses the cross-cutting demand for seasonal risk awareness, baseline comparisons, and pattern recognition. Supports advisors' seasonal planning, producers' harvest monitoring, policy actors' trend detection, distributors' sourcing timing, and technicians' campaign-level risk assessment. **Comment** can the chart be simplified?
- C6 – Geographic Heatmap: The most directly requested chart type across all stakeholder groups. Policy actors explicitly requested country-level colour-coded maps (DR1), advisors wanted location-aware risk displays, producers need localised risk, distributors need regional risk classification, and technicians requested geospatial/satellite mapping. **Comment** we suggest this chart to be the first one on the dashboard and to be adjusted on the different climatic scenarios (good, most likely to happened and the bad one)

Additionally, C1 – Toxin Concentration vs. Time appears as Primary in five of six views (Secondary only for Policy Actors), reflecting the universal focus on mycotoxin monitoring across all stakeholder categories. **Comment** that chart should change accordingly, if the user chooses an enteric pathogen this chart **should NOT** pop-up.

2.2 View-Specific Emphasis

- C4 (Cases per 100k vs. Time) is Primary only in the Policy Actor view and excluded from Producer and Distributor views. This reflects the epidemiological orientation of policy actors (risk classification, outbreak monitoring) versus the operational focus of producers and distributors, who care about contamination levels rather than public health statistics.
- C7 (Climate-Adjusted Scenarios vs. Time) is Primary for Advisors, Policy Actors, and Technicians the three groups that explicitly requested climate integration, predictive modelling, and scenario analysis. It is excluded from the Distributor view, where the simpler, action-oriented focus does not require complex scenario exploration.
- C2 (Pathogen Concentration 3D) is Primary for Advisors, Producers **Comment** (NOT the 3D one, we conceder that to be very difficult for them to understand), and Technicians groups that need granular pathogen data for field-level and crop-specific decisions. It is Secondary for Distributors and Policy Actors, who need the information but at a more aggregated level.
- C3 (Probability of Illness vs. Time) is Primary for Advisors, Policy Actors, and Technicians, reflecting their emphasis on risk prediction and probability of exceedances. It is Secondary for Producers and Distributors, who use it as supplementary context rather than a primary decision driver.

3. Per-View Chart Representation

The same chart should be rendered differently depending on the selected view. The stakeholder feedback strongly supports view-specific configuration across five parameter dimensions. Below we define each dimension, justify it with stakeholder feedback, specify the defaults per view, and state whether users can override them.

3.1 Time Range

Stakeholder justification: Advisors showed a tension between long-term and in-season utility (DR2). Producers need short-term operational windows linked to harvest monitoring (DR1). Policy actors focus on multi-season trend detection and long-term scenario analysis (DR2, DR5). Technicians work at campaign/season level with precise forecasting (DR3, DR5). Distributors focus on upcoming sourcing windows (DR2, DR3). **Comment** for sure those ranges should be modified according to our models.

View	Default Time Range	Min Selectable	Max Selectable	User Override
Default	Current season	1 month	5 years	Yes
Advisor	Current + next season	1 month	5 years	Yes
Policy Actor	Last 12 months	3 months	10 years	Yes
Producer	Current season	1 week	2 years	Yes
Distributor	Next 3 months	1 month	1 year	Yes
Technician	Current campaign	1 week	3 years	Yes

3.2 Geographic Scope

Stakeholder justification: Policy actors want country-level and regional overviews (DR1). Advisors need multi-region comparison with local drill-down (DR2). Producers emphasise highly localised risk (DR2, Q3). Distributors need footprints smaller than broad regions (DR2). Technicians want field-to-local scope with geospatial precision (DR3, DR5).

View	Default Scope	Drill-Down To	Aggregation	User Override
Default	Country/region	Sub-region	Regional	Yes
Advisor	Region (multi-select)	Local/field	Regional	Yes
Policy Actor	National/EU-wide	Sub-region	Aggregated	Yes
Producer	Local (user location)	Field-level	None	Yes
Distributor	Sourcing regions	Sub-region	Regional	Yes

Technician	Local/campaign area	Field-level	None	Yes

3.3 Level of Detail

Stakeholder justification: Producers explicitly requested granular statistical data (median, mean, SD, outliers) (DR1) **Comment** that feedback came from only one DR, we suggest that for the farmer the dashboard should be as easy and understandable as possible, statistics can be overwhelming. Technicians want quantified risk indicators (probabilities, crop losses) (DR5). Policy actors prefer summary-level aggregated outputs for quick orientation (DR1, DR4). Distributors favour simplified, action-oriented displays (DR2). Advisors need both summary and detailed modes depending on the advisory task (DR2, DR4).

View	Default Detail Level	Available Modes	Statistical Display
Default	Summary	Summary, Detailed	Mean only
Advisor	Summary + Uncertainty	Summary, Detailed, Export	Mean, Confidence Interval (CI), exceedance prob.
Policy Actor	Aggregated summary	Summary, Trend overlay	Mean, trend lines
Producer	Detailed breakdown	Summary, Detailed, Raw	Mean, median, SD, outliers
Distributor	Summary	Summary only	Mean, thresholds
Technician	Detailed + Alerts	Summary, Detailed, Alert	Mean, CI, probabilities

User override: Yes for all views. Users can switch between available detail modes within their view.

3.4 Default Filters

Stakeholder justification: All groups requested that the platform tailor outputs based on profile data (crop, region, hazard). Producers need crop-specific and supply-chain-aware filtering (DR1, Q2). Technicians distinguish between human food and animal feed (DR3, Q3). Policy actors want multi-hazard, multi-crop views (DR2, DR5). Advisors want location-aware and climate-integrated filtering (DR4, DR5). **Comment** all those extra filters can be conceder in the next version depenting on the platform capabilities.

View	Crop Filter	Hazard Filter	Region Filter	Risk Level	Override
Default	From profile	From profile	From profile	All	Yes
Advisor	Multi-select	Multi-select	Multi-region	All	Yes
Policy Actor	All crops	Multi-hazard	National	All	Yes
Producer	Profile crop	Profile hazard	Profile loc.	All	Yes
Distributor	Commodity	Profile hazard	Sourcing area	High	Yes
Technician	Profile crop	Profile hazard	Local	All	Yes

All filters are user-overridable. The defaults represent the most useful starting configuration based on stakeholder feedback, but users should never be locked into a fixed filter set. This addresses the flexibility concerns raised by Advisors (DR4: “profile should allow multiple selections”) and Policy Actors (DR2: “flexibility for complex user roles”).

3.5 Visual Emphasis & Layout

Stakeholder justification: Technicians requested traffic-light colour-coded alerts (DR3). Policy actors want quick risk overviews with colour intensity (DR1). Distributors need action-oriented outputs (DR2). Producers want real-time alert emphasis (DR5). Advisors requested uncertainty and reliability indicators (DR2). **Comment** we suggest the traffic-light colour-coded alerts to be used across all user types.

View	Key Metric Highlighted	Secondary (Hidden by Default)	Special Visual Treatment
Default	Risk level indicator	Statistical details	Standard chart layout
Advisor	Exceedance probability	Raw concentration values	Uncertainty bands on all time-series
Policy Actor	Trend direction + Cases	Concentration details	Trend arrows, aggregated overlays
Producer	Current concentration	Illness probability	Alert banners for threshold breaches
Distributor	Risk zone classification	Pathogen detail	Simplified risk badges (High/Med/Low)

Technician	Predicted risk level	Epidemiological metrics	Traffic-light indicators on all charts

User override: Partial. Users can expand hidden secondary metrics, but the primary highlighted metric and visual treatment (traffic lights, trend arrows, etc.) are fixed per view to maintain conceptual clarity.

4. End-to-End Examples

As requested, below are three end-to-end examples showing how the same chart appears differently across views, with specific parameter changes documented.

Example 1: C6 – Geographic Heatmap

This chart is Primary across all views. It illustrates the widest range of per-view variation.

In the Policy Actor View

- Geographic scope: National or EU-wide. The map defaults to showing an entire country or group of countries, with sub-regional colour coding.
- Time range: Last 12 months (aggregated). Risk intensity reflects the cumulative or average risk over the past year.
- Default filter: All crops, all hazards. Policy actors need the multi-risk perspective (DR5) to perform cross-hazard comparisons.
- Level of detail: Aggregated summary. Each region shows a single composite risk score.
- Visual emphasis: Colour-coded intensity bands at national/regional resolution. Trend arrows overlaid to indicate whether risk is increasing, stable, or decreasing versus the previous period.
- User override: Can drill down to sub-region. Can switch to single-hazard filter. Cannot remove trend arrows (fixed visual treatment).

In the Producer View

- Geographic scope: Local (centred on user's registered location). The map defaults to a tight radius around the user's farm or production site.
- Time range: Current season. Risk intensity reflects the live/recent monitoring data for the active growing or harvest window.
- Default filter: Profile crop and profile hazard only. The map is pre-filtered to show only what is directly relevant to the user's operation.
- Level of detail: Detailed breakdown. Each area can be expanded to show mean, median, and threshold exceedance statistics.
- Visual emphasis: Alert banners overlaid on high-risk zones. Threshold breach zones are highlighted with pulsing or bold outlines. The primary metric shown is current toxin concentration relative to regulatory limits.
- User override: Can zoom out to regional level. Can add additional crops/hazards to filter. Cannot remove alert banners.

In the Technician View

- Geographic scope: Local/campaign area. Defaults to the geographic scope of the technician's active advisory campaign.
- Time range: Current campaign (aligned with growing season).

- Default filter: Profile crop, profile hazard, local area.
- Level of detail: Detailed with quantified risk indicators per zone.
- Visual emphasis: Traffic-light overlay on each geographic zone (red/yellow/green). Zones are labelled with predicted risk probability percentage. Satellite/environmental base layer enabled by default.
- User override: Can switch to broader region. Can toggle satellite layer off. Traffic-light system is fixed.

Example 2: C1 – Toxin Concentration vs. Time

This chart is Primary in five views and Secondary for Policy Actors. It shows how time range, detail level, and visual emphasis shift by role.

In the Advisor View

- Time range: Current + next season (~6–9 months forward). Advisors need to anticipate upcoming risks (DR3).
- Geographic scope: Multi-region selectable. Advisors often compare risk across the regions they cover.
- Level of detail: Summary with uncertainty. The line chart shows the central estimate with shaded confidence intervals. Exceedance probability thresholds are overlaid as horizontal reference lines.
- Default filter: Multi-crop selectable, multi-hazard. Advisors work across diverse farms and crops (DR4).
- Visual emphasis: Uncertainty bands are the primary visual feature. Where the confidence interval crosses regulatory thresholds, the area is highlighted.
- User override: Full. Can switch to raw data mode, change crops, export data (CSV/Excel as requested DR4).

In the Producer View

- Time range: Current season (~3 months). Operational focus on the active harvest window (DR1).
- Geographic scope: Single location (user's site). No multi-region comparison by default.
- Level of detail: Detailed breakdown. The chart includes the mean line plus individual measurement points. Statistical panel below the chart shows median, standard deviation, and outlier flags—directly matching the DR1 request for granular statistical data.
- Default filter: Profile crop and hazard. One crop, one hazard, one location.
- Visual emphasis: Regulatory threshold line is bold and colour-coded (red). When the current or forecast concentration approaches the threshold, an alert banner appears above the chart. This addresses the real-time alert need (DR5).
- User override: Can extend time range. Can add additional data series. Alert banner is fixed.

In the Distributor View

- Time range: Next 3 months (upcoming sourcing window).
- Geographic scope: Sourcing regions (pre-filtered to the distributor's supply chain geography).
- Level of detail: Summary only. Clean line with mean values and clear threshold markers.
- Default filter: Commodity-level crop, relevant hazards. Filtered to commercially relevant products.
- Visual emphasis: Simplified risk badge (High/Medium/Low) displayed alongside the chart title based on the forecast trajectory. The chart prioritises actionability—the distributor can quickly determine whether sourcing from a region is safe. This addresses the action-oriented output demand (DR2).
- User override: Can switch regions. Cannot access raw statistical breakdown (this would need Advisor or Producer view).

Example 3: C7 – Climate-Adjusted Scenarios vs. Time

This chart is Primary for Advisors, Policy Actors, and Technicians; Secondary for Producers; and excluded from the Distributor view.

In the Policy Actor View

- Time range: Multi-year (3–10 years). Policy actors need long-term trend detection for strategic planning (DR2).
- Geographic scope: National/EU-wide. Scenario projections aggregated at the national level.
- Level of detail: Aggregated summary with scenario bands. Multiple climate scenarios (e.g., RCP 4.5, RCP 8.5) displayed as diverging projection lines.
- Default filter: All crops, all hazards—multi-risk perspective. Policy actors want to understand how climate change affects the entire risk landscape.
- Visual emphasis: Scenario comparison is the primary feature. Divergence points between scenarios are highlighted. Trend direction indicators show acceleration or deceleration of risk. This directly supports the forecasting and trend detection requested (DR2, DR5).
- User override: Can select individual scenarios. Can filter to specific crops/hazards. Scenario comparison layout is fixed.

In the Technician View

- Time range: Current campaign to next campaign (~6–12 months). Technicians need precise near-term climate forecasting for proactive adaptation (DR3).
- Geographic scope: Local/campaign area. Scenarios projected at the resolution relevant to the technician's advisory territory.
- Level of detail: Detailed with quantified impact estimates. Each scenario line shows not only the risk trajectory but also the estimated crop loss probability associated with it—addressing the risk-yield integration request (DR3).
- Default filter: Profile crop and hazard. Single-crop, single-hazard focus for precision.
- Visual emphasis: Traffic-light zones projected forward along the time axis. Green/yellow/red bands on the scenario chart show when risk is expected to cross management thresholds. This supports the technician's need to advise on timing decisions (earlier sowing, variety selection). Decision points are marked on the chart.
- User override: Can extend time horizon. Can overlay additional crops. Traffic-light zones are fixed.

In the Advisor View

- Time range: Current + next season (medium-term).
- Geographic scope: Multi-region. Advisors compare how climate scenarios affect different regions they cover.
- Level of detail: Summary with uncertainty bands per scenario. Each scenario shows a central estimate plus confidence interval.
- Default filter: Multi-crop, relevant hazards. Reflects the advisor's cross-farm, cross-region advisory scope.
- Visual emphasis: Uncertainty is the primary visual layer. Where scenario confidence intervals overlap heavily, the chart indicates low differentiation between scenarios. Where they diverge, the chart

highlights the decision-sensitive window. This supports the contextualised insight combining quantitative data with uncertainty that advisors requested (DR2, DR5).

- User override: Full flexibility to filter, export, and switch between scenarios.

5. User Override Policy

A clear distinction emerged from the stakeholder feedback between parameters that should be user-adjustable and those that should remain view-fixed to preserve the conceptual integrity of each mode.

5.1 Always User-Overridable

- Time range (within view-defined bounds): All groups expressed different temporal needs within their role, and even within a single role the task at hand varies. Fixed time ranges would frustrate all user types.
- Geographic scope (drill up/down): Every group requested the ability to explore at different geographic levels. The default sets the most useful starting point, but exploration must be unrestricted.
- Filters (crop, hazard, region, risk level): Advisors (DR4) and Policy Actors (DR2) explicitly requested multi-selection flexibility. No filter should be permanently locked.
- Detail level (within available modes): Producers and Technicians need detailed modes, but should be able to switch to summary when doing a quick scan.

5.2 Fixed Per View (Not User-Overridable)

- Visual treatment (traffic lights, trend arrows, alert banners, uncertainty bands, risk badges): These are the defining identity of each view. Changing them would effectively switch the user to a different view. If a user wants a different visual treatment, they should switch views.
- Primary highlighted metric: Tied to the visual treatment. Changing it would break the conceptual coherence of the view.
- Chart priority and order: The layout hierarchy reflects the role's workflow. A technician should not have to manually re-arrange charts to get an advisor layout they should switch to the Advisor view.
- Available detail modes: Some views intentionally exclude raw/export modes (e.g., Distributor) to maintain simplicity. Users needing those capabilities should use a more detailed view.
- Time range bounds: The minimum and maximum selectable ranges enforce the conceptual scope of each view (e.g., Distributors should not be plotting 10-year climate scenarios).